

ARPANA SINGH

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SUMMARY

Passionate Data Engineer, having 10 years of overall IT experience. Proficient in software development, client management and execution of software projects. Experienced in complete project life cycle from analysis, architecture, design, development, deployment & support. Accomplished successful lead initiatives to drive sales and customer satisfaction. Active participant of hiring, mentoring teams. Ability to see the business aspect and technical aspect of a project to best suit the needs of every project and employer.

EDUCATION

San Jose State University, San Jose (Jan 2025 – Present)

- MS, Applied Data Intelligence

Cornell University, NY (Jan 2023 – June 2023)

- Bootcamp Training – Python for Data Science

Cochin University of Science & Technology, India (June 2011)

- B.Tech., Civil Engineering

SKILLS

Programming Languages: Python, SQL, Java

Data Science Tools: Airflow, Kafka, Spark, Hadoop.

Operating Systems: Linux / Unix, Windows, MacOS

Cloud Platform: AWS, GCP

Databases: Neo4j, MongoDB, Postgres, Oracle

ETL Tools: Glue, Informatica, ELK Stack

Search Engines: Elasticsearch, Kibana

CI / CD Tools: Jenkins, Jira, Confluence

Container Technologies: Docker

Hadoop Ecosystem: HDFS, Hive

ACADEMIC PROJECTS:

Student, San Jose State University, San Jose, CA

Jan 2025 – Present

- Implemented PySpark ETL pipeline to process the allcuration research data from amazon marketplace. Real-time data processing and integration with cloud services using Google Managed Kafka (Cloud Pub/Sub for Apache Kafka)
- Implemented Big Data database creation with Python, MongoDB MapReduce, and Docker.
- Implement a PySpark program to list out the Wikipedia pages with the top ten PageRanks on a sample Wikipedia XML dataset.
- Implemented data workflow fetching Twitter data and ingesting into GCP Big Query table using Airflow, Kafka Streaming; Elasticsearch to index the table data, Kibana to visualize trending hashtags.
- Designed and developed Redshift AWS data warehouse, providing real-time analytics comparing healthcare insurance data across US & EU.
- Dynamic dashboard creation using Power BI & Tableau using olympic 2024 data to showcase the potential gender parity, medal tally across different countries/sports and historical transition over the years.

Data Science Trainee, Cornell University, NY

Jan 2023 – June 2023

- Conducted in-depth analysis of the World Happiness Report dataset, employing Pandas & Matplotlib establishing correlations among variables (such as health, freedom, and trust).
- Utilized the CDC's COVID disease dataset to extract and engineer relevant features that significantly impacted the number of casualties.
- Designed and fine-tuned a neural network model with three hidden layers, optimizing hyperparameters such as the number of hidden layers, input features, and training epochs. Achieved an impressive 88.7% accuracy on the COVID disease dataset through iterative refinement.

WORK EXPERIENCE:

Data Engineer Lead, Capgemini, Bengaluru, India

Aug 2019 – Oct 2021

- Spearheaded the first cloud migration project for Warner Bros, from on-premises solutions to AWS, encompassing requirement gathering, design, implementation, and validation.
- Implemented migration of postgres database application from on-premise to cloud.
- Notably improved client satisfaction, contributing to a 15% vertical growth in 2020.

Data Engineer Lead, Capgemini, Toronto, Canada

Aug 2018 – Jul 2019

- Collaborated with client (Rogers Communications) on data warehouse (DWH) projects design and solutions review.
- Designed and executed SQL scripts using Informatica for data flow and transformation, while also overseeing solution integration and expediting bug fixes.

Data Engineer Lead, Capgemini, Bengaluru, India

Dec 2016 – Jul 2018

- Led ETL Dev team, fostering collaboration through design and code reviews, and streamlined processes facilitating multiple projects.
- Implemented performance tuning using Pushdown Optimization (PDO) and partitioning.

Senior Software Engineer / ETL Lead, UST Global, Kochi, India

Jan 2015 – Nov 2016

- Designed and developed Informatica mappings, sessions, and workflows, improving data extraction from multiple source systems for efficient reporting in the data warehouse.
- Providing cross-functional support to other teams to resolve production issues efficiently.

Software Engineer, UST Global, Kochi, India

Jan 2012 – Dec 2014

- Designed and developed data extraction and load using Informatica.
- Prepared test documents and worked on additional technical documentation outlining the entire ETL process for our client, Anthem Blue Cross Blue Shield.
- Reviewed development codes of my peers to ensure the requirements of our client, and our high standard of our delivery have been met.
- Automated the integration and stress testing for our project in order to support our scalable solutions.